Math 39 Curriculum

Precalculus with Data Analysis | Middlesex School Mathematics

COURSE AT A GLANCE

Linear Modeling and Residuals

REQUIRED

Supplemental resources: 3.3 Rates of Change and Behavior of Graphs; 4.1 Linear Functions; 4.3 Fitting Linear Models to Data

UNIT LEARNING OBJECTIVES

I can calculate and interpret average rate of change with appropriate units.

I can construct a linear function from two data points, a table, a graph, or a context.

I can fit a linear regression model to data using technology and interpret its parameters.

I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.

I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.

I can use a linear model to make and interpret predictions, including solving for either variable.

Exponential Modeling

REQUIRED

Supplemental resources: College Algebra 2e 6.1 Exponential Functions; Precalculus 2e 4.1 Exponential Functions; Precalculus 2e 4.7 Exponential and Logarithmic Models; Precalculus 2e 4.6 Exponential and Logarithmic Equations; Precalculus 2e 4.8 Fitting Exponential Models to Data; College Algebra 2e 4.3 Fitting Linear Models to Data

UNIT LEARNING OBJECTIVES

I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.

I can construct an exponential function from two data points, a table, or a context.

I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.

I can determine and interpret doubling time or half-life from an exponential model.

I can solve exponential and logarithmic equations in modeling contexts.

I can fit an exponential regression model using technology and evaluate its fit.

I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Power Functions and Function Comparison

REQUIRED

Supplemental resources: Precalculus 2e 3.3 Power Functions and Polynomial Functions; Precalculus 2e 3.9 Modeling Using Variation; College Algebra 2e 2.6 Other Types of Equations; Precalculus 2e 4.2 Graphs of Exponential Functions; Precalculus 2e 4.4 Graphs of Logarithmic Functions

UNIT LEARNING OBJECTIVES

I can identify a power function and interpret the parameters in y = kx^p.

I can construct a power function from two data points or a proportional relationship.

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

I can recognize and model direct, inverse, and other power-variation relationships.

I can solve equations involving integer or rational powers.

I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.

I can fit a power regression model and interpret its parameters and correlation evidence.

Function Combinations and Polynomials

REQUIRED

Supplemental resources: 1.4 Composition of Functions; 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions

UNIT LEARNING OBJECTIVES

I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.

I can determine where a combination of functions is positive, zero, or undefined.

I can determine polynomial end behavior from degree and leading coefficient.

I can identify zeros and multiplicities and relate them to polynomial graph behavior.

I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.

I can construct a polynomial function from specified zeros, multiplicities, and a point.

I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.

I can build and optimize a polynomial model in a geometric or economic context.

Describing Data

REQUIRED

No mapped OpenStax section

UNIT LEARNING OBJECTIVES

I can distinguish categorical and quantitative data.

I can construct and interpret frequency and relative-frequency tables.

I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.

I can identify misleading features in a data display.

I can describe a distribution's modality, symmetry, skewness, and outliers.

I can calculate and interpret mean, median, and mode and select an appropriate measure of center.

I can calculate and interpret range, standard deviation, quartiles, and interquartile range.

I can construct and interpret a five-number summary and boxplot.

I can use percentiles and z-scores to locate and compare values within distributions.

I can compare distributions using their shape, center, spread, and unusual features.

Probability

REQUIRED

Supplemental resources: 11.7 Probability

UNIT LEARNING OBJECTIVES

I can interpret and write notation for compound and conditional probabilities.

I can organize categorical data in a contingency table or Venn diagram.

I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.

I can identify an experiment, outcomes, events, and a sample space.

I can calculate theoretical probabilities when outcomes are equally likely.

I can use complements to calculate probabilities.

I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.

I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.

I can compare experimental and theoretical probability and explain the law of large numbers.

Expected Value

EXTENSION

No mapped OpenStax section

UNIT LEARNING OBJECTIVES

I can calculate the expected value of a discrete probability situation.

I can interpret expected value to evaluate a game, risk, or long-run decision.

Linearization of Nonlinear Data

EXTENSION

No mapped OpenStax section

UNIT LEARNING OBJECTIVES

I can linearize power or exponential data and translate between the linearized and original models.

IN-DEPTH CURRICULUM

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Linear Modeling and Residuals

REQUIRED

OpenStax coverage: **Supplemental resources:** 3.3 Rates of Change and Behavior of Graphs; 4.1 Linear Functions; 4.3 Fitting Linear Models to Data

M39-LIN-001

I can calculate and interpret average rate of change with appropriate units.

OpenStax: **Supplemental resources:** 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

- D Calculate average rate of change over an interval.
  SK-FUNC-AROC · first introduced in Math 21
- D Interpret average rate of change with contextual units.
  SK-FUNC-AROC-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M39-LIN-002

I can construct a linear function from two data points, a table, a graph, or a context.

OpenStax: **Supplemental resources:** 4.1 Linear Functions

SUPPORTING SKILLS

- R Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12
- A Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12
- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M39-LIN-003

I can fit a linear regression model to data using technology and interpret its parameters.

OpenStax: **Supplemental resources:** 4.3 Fitting Linear Models to Data

SUPPORTING SKILLS

- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- I Use technology to determine a least-squares linear regression model.
  SK-REG-LINEAR-FIT
- I Recall and correctly use regression terms including explanatory variable, response variable, correlation, residual, and residual plot.
  SK-REG-VOCABULARY

M39-LIN-004

I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.

OpenStax: **Supplemental resources:** 4.3 Fitting Linear Models to Data

SUPPORTING SKILLS

- I Interpret the sign and magnitude of a correlation coefficient in context.
  SK-REG-CORRELATION-INTERPRET

M39-LIN-005

I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.

OpenStax: **Supplemental resources:** 4.3 Fitting Linear Models to Data

SUPPORTING SKILLS

- I Use residual patterns and contextual evidence to judge whether a model is appropriate.
  SK-REG-MODEL-ASSESS
- I Calculate an observed value minus its model-predicted value.
  SK-REG-RESIDUAL-CALCULATE
- I Construct and interpret a residual plot.
  SK-REG-RESIDUAL-PLOT

M39-LIN-006

I can use a linear model to make and interpret predictions, including solving for either variable.

OpenStax: **Supplemental resources:** 4.3 Fitting Linear Models to Data

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Solve for an input associated with a specified output.
  SK-FUNC-SOLVE-INPUT · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12

Exponential Modeling

REQUIRED

OpenStax coverage: **Supplemental resources:** College Algebra 2e 6.1 Exponential Functions; Precalculus 2e 4.1 Exponential Functions; Precalculus 2e 4.7 Exponential and Logarithmic Models; Precalculus 2e 4.6 Exponential and Logarithmic Equations; Precalculus 2e 4.8 Fitting Exponential Models to Data; College Algebra 2e 4.3 Fitting Linear Models to Data

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M39-EXP-001

I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.

OpenStax: **Supplemental resources:** 6.1 Exponential Functions

SUPPORTING SKILLS

- D Use constant differences or ratios to distinguish linear and exponential patterns.
  SK-EXP-DIFFERENCE-RATIO · first introduced in Math 31
- D Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21

M39-EXP-002

I can construct an exponential function from two data points, a table, or a context.

OpenStax: **Supplemental resources:** 4.1 Exponential Functions

SUPPORTING SKILLS

- D Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- D Determine an exponential model from tabular values.
  SK-EXP-MODEL-TABLE · first introduced in Math 31
- D Solve for an exponential model using two points.
  SK-EXP-MODEL-TWO-POINTS · first introduced in Math 31

M39-EXP-003

I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.

OpenStax: **Supplemental resources:** 4.1 Exponential Functions

SUPPORTING SKILLS

- D Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- R Recall and correctly use exponential-function terms including initial value, growth or decay factor, growth or decay rate, and asymptote.
  SK-EXP-VOCABULARY · first introduced in Math 31
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M39-EXP-004

I can determine and interpret doubling time or half-life from an exponential model.

OpenStax: **Supplemental resources:** 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- A Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- D Translate a doubling-time or half-life question into an exponential equation.
  SK-LOG-DOUBLING-HALF · first introduced in Math 31

M39-EXP-005

I can solve exponential and logarithmic equations in modeling contexts.

OpenStax: **Supplemental resources:** 4.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

- A Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- R Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- R Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG · first introduced in Math 31

M39-EXP-006

I can fit an exponential regression model using technology and evaluate its fit.

OpenStax: **Supplemental resources:** 4.8 Fitting Exponential Models to Data

SUPPORTING SKILLS

- A Interpret the sign and magnitude of a correlation coefficient in context.
  SK-REG-CORRELATION-INTERPRET · first introduced in Math 39
- I Use technology to determine an exponential regression model.
  SK-REG-EXPONENTIAL-FIT
- D Use residual patterns and contextual evidence to judge whether a model is appropriate.
  SK-REG-MODEL-ASSESS · first introduced in Math 39

M39-EXP-007

I can compare linear and exponential models and justify which is more appropriate for a data set or context.

OpenStax: **Supplemental resources:** Precalculus 2e 4.7 Exponential and Logarithmic Models; College Algebra 2e 4.3 Fitting Linear Models to Data; Precalculus 2e 4.8 Fitting Exponential Models to Data

SUPPORTING SKILLS

- A Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- I Select a function family whose behavior fits a table, graph, or context.
  SK-MODEL-FAMILY-SELECT
- D Use residual patterns and contextual evidence to judge whether a model is appropriate.
  SK-REG-MODEL-ASSESS · first introduced in Math 39

Power Functions and Function Comparison REQUIRED

OpenStax coverage: **Supplemental resources:** Precalculus 2e 3.3 Power Functions and Polynomial Functions; Precalculus 2e 3.9 Modeling Using Variation; College Algebra 2e 2.6 Other Types of Equations; Precalculus 2e 4.2 Graphs of Exponential Functions; Precalculus 2e 4.4 Graphs of Logarithmic Functions

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M39-POW-001

I can identify a power function and interpret the parameters in y = kx^p.

OpenStax: **Supplemental resources:** 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

- I — Interpret the coefficient and exponent in a power function.
  SK-POWER-PARAMETER-INTERPRET
- I — Recall and correctly use power-function terms including coefficient, exponent, direct variation, and inverse variation.
  SK-POWER-VOCABULARY
- I — Represent and interpret a power-variation relationship.
  SK-VAR-POWER

M39-POW-002

I can construct a power function from two data points or a proportional relationship.

OpenStax: **Supplemental resources:** 3.9 Modeling Using Variation

SUPPORTING SKILLS

- A — Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I — Determine a power function from two points or a proportional relationship.
  SK-POWER-CONSTRUCT

M39-POW-003

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

OpenStax: **Supplemental resources:** 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

- I — Determine power-function behavior as the input approaches zero or positive or negative infinity.
  SK-POWER-END-BEHAVIOR
- I — Relate the coefficient and exponent of a power function to its graph shape.
  SK-POWER-SHAPE
- D — Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M39-POW-004

I can recognize and model direct, inverse, and other power-variation relationships.

OpenStax: **Supplemental resources:** 3.9 Modeling Using Variation

SUPPORTING SKILLS

- I — Represent and interpret a direct-variation relationship.
  SK-VAR-DIRECT
- I — Represent and interpret an inverse-variation relationship.
  SK-VAR-INVERSE
- D — Represent and interpret a power-variation relationship.
  SK-VAR-POWER · first introduced in Math 39

M39-POW-005

I can solve equations involving integer or rational powers.

OpenStax: **Supplemental resources:** 2.6 Other Types of Equations

SUPPORTING SKILLS

- **A** Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- **A** Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21

M39-POW-006

I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.

OpenStax: **Supplemental resources:** 3.3 Power Functions and Polynomial Functions; 4.2 Graphs of Exponential Functions; 4.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

- **D** Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- **I** Compare the long-run growth or decay of functions from different families.
  SK-FUNC-DOMINANCE

M39-POW-007

I can fit a power regression model and interpret its parameters and correlation evidence.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- **D** Interpret the coefficient and exponent in a power function.
  SK-POWER-PARAMETER-INTERPRET · first introduced in Math 39
- **A** Interpret the sign and magnitude of a correlation coefficient in context.
  SK-REG-CORRELATION-INTERPRET · first introduced in Math 39
- **I** Use technology to determine a power regression model.
  SK-REG-POWER-FIT

Function Combinations and Polynomials REQUIRED

OpenStax coverage: **Supplemental resources:** 1.4 Composition of Functions; 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions

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M39-POL-001

I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.

OpenStax: **Supplemental resources:** 1.4 Composition of Functions

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I Add, subtract, multiply, and divide function outputs.
  SK-FUNC-OPERATIONS

M39-POL-002

I can determine where a combination of functions is positive, zero, or undefined.

OpenStax: **Supplemental resources:** 1.4 Composition of Functions; 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

- I Determine the domain of a sum, difference, product, quotient, or composition.
  SK-FUNC-DOMAIN-COMBINE
- D Add, subtract, multiply, and divide function outputs.
  SK-FUNC-OPERATIONS · first introduced in Math 39

M39-POL-003

I can determine polynomial end behavior from degree and leading coefficient.

OpenStax: **Supplemental resources:** 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

- I Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR
- R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M39-POL-004

I can identify zeros and multiplicities and relate them to polynomial graph behavior.

OpenStax: **Supplemental resources:** 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

- I Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12
- D Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M39-POL-005

I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.

OpenStax: **Supplemental resources:** 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

- I — Use inflection points to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-INFLECTION
- I — Use turning points to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-TURNING
- I — Use zeros and their multiplicities to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-ZEROS
- D — Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR · first introduced in Math 39
- D — Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39

M39-POL-006

I can construct a polynomial function from specified zeros, multiplicities, and a point.

OpenStax: **Supplemental resources:** 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

- D — Build a polynomial from zeros, multiplicities, and a point.
  SK-POLY-CONSTRUCT · first introduced in Math 31
- A — Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39
- A — Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M39-POL-007

I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.

OpenStax: **Supplemental resources:** 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

- D — Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR · first introduced in Math 39
- A — Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39
- A — Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M39-POL-008

I can build and optimize a polynomial model in a geometric or economic context.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- A — Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A — Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D — Interpret vertex and zeros in an application.
  SK-QUAD-MODEL-FEATURES · first introduced in Math 21

Describing Data

REQUIRED

OpenStax coverage: No mapped OpenStax section

M39-STA-001

I can distinguish categorical and quantitative data.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Classify data as categorical or quantitative.
  SK-STAT-DATA-TYPE

M39-STA-002

I can construct and interpret frequency and relative-frequency tables.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Construct and interpret a frequency table.
  SK-STAT-FREQUENCY-TABLE
- I Calculate and interpret relative frequency.
  SK-STAT-RELATIVE-FREQUENCY

M39-STA-003

I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Construct and interpret bar charts and pie charts.
  SK-STAT-CATEGORICAL-DISPLAY
- I Construct and interpret a histogram for quantitative data.
  SK-STAT-HISTOGRAM

M39-STA-004

I can identify misleading features in a data display.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Identify perceptual distortion, misleading scales, and other misleading features in data displays.
  SK-STAT-GRAPH-CRITIQUE

M39-STA-005

I can describe a distribution's modality, symmetry, skewness, and outliers.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Describe a distribution by modality, symmetry, and skewness.
  SK-STAT-DISTRIBUTION-SHAPE
- I Identify and interpret unusual observations or outliers.
  SK-STAT-OUTLIER-IDENTIFY
- I Recall and correctly use terms for data type, distribution shape, center, spread, percentile, and outlier.
  SK-STAT-VOCABULARY

M39-STA-006

I can calculate and interpret mean, median, and mode and select an appropriate measure of center.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Select an appropriate measure of center based on distribution shape and context.
  SK-STAT-CENTER-SELECT
- I Calculate and interpret the arithmetic mean.
  SK-STAT-MEAN
- I Calculate and interpret the median.
  SK-STAT-MEDIAN
- I Identify and interpret the mode or modes.
  SK-STAT-MODE

M39-STA-007

I can calculate and interpret range, standard deviation, quartiles, and interquartile range.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Determine quartiles and calculate and interpret the interquartile range.
  SK-STAT-QUARTILES-IQR
- I Calculate and interpret the range.
  SK-STAT-RANGE
- I Calculate and interpret standard deviation as a measure of spread.
  SK-STAT-STANDARD-DEVIATION

M39-STA-008

I can construct and interpret a five-number summary and boxplot.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Construct and interpret a boxplot.
  SK-STAT-BOXPLOT
- I Determine and interpret a five-number summary.
  SK-STAT-FIVE-NUMBER

M39-STA-009

I can use percentiles and z-scores to locate and compare values within distributions.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Determine and interpret the percentile rank of a value.
  SK-STAT-PERCENTILE
- I Calculate and interpret a z-score.
  SK-STAT-ZSCORE

M39-STA-010

I can compare distributions using their shape, center, spread, and unusual features.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- A Select an appropriate measure of center based on distribution shape and context.
  SK-STAT-CENTER-SELECT · first introduced in Math 39
- I Compare distributions using shape, center, spread, and unusual features.
  SK-STAT-DISTRIBUTION-COMPARE
- A Describe a distribution by modality, symmetry, and skewness.
  SK-STAT-DISTRIBUTION-SHAPE · first introduced in Math 39

Probability

REQUIRED

OpenStax coverage: **Supplemental resources:** 11.7 Probability

M39-PRO-009

I can interpret and write notation for compound and conditional probabilities.

OpenStax: **Supplemental resources:** 11.7 Probability

SUPPORTING SKILLS

- I Read and write probability notation and correctly use the terms outcome, event, sample space, complement, intersection, union, and conditional probability.
  SK-PROB-NOTATION-VOCAB

M39-PRO-001

I can organize categorical data in a contingency table or Venn diagram.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Construct and interpret a contingency table.
  SK-PROB-CONTINGENCY-TABLE
- I Represent categorical events and their overlap with a Venn diagram.
  SK-PROB-VENN-DIAGRAM

M39-PRO-002

I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Calculate and interpret a conditional probability.
  SK-PROB-CONDITIONAL
- I Calculate probability from observed relative frequency.
  SK-PROB-EMPIRICAL
- I Calculate and interpret marginal and joint probabilities.
  SK-PROB-MARGINAL-JOINT

M39-PRO-003

I can identify an experiment, outcomes, events, and a sample space.

OpenStax: **Supplemental resources:** 11.7 Probability

SUPPORTING SKILLS

- A Read and write probability notation and correctly use the terms outcome, event, sample space, complement, intersection, union, and conditional probability.
  SK-PROB-NOTATION-VOCAB · first introduced in Math 39
- I Identify outcomes, events, and the sample space of an experiment.
  SK-PROB-SAMPLE-SPACE

M39-PRO-004

I can calculate theoretical probabilities when outcomes are equally likely.

OpenStax: **Supplemental resources:** 11.7 Probability

SUPPORTING SKILLS

- A Identify outcomes, events, and the sample space of an experiment.
  SK-PROB-SAMPLE-SPACE · first introduced in Math 39
- I Calculate theoretical probability for equally likely outcomes.
  SK-PROB-THEORETICAL

M39-PRO-005

I can use complements to calculate probabilities.

OpenStax: **Supplemental resources:** 11.7 Probability

SUPPORTING SKILLS

- I Use the complement rule to calculate probability.
  SK-PROB-COMPLEMENT

M39-PRO-006

I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Calculate the probability of a compound and event.
  SK-PROB-AND
- D Calculate and interpret a conditional probability.
  SK-PROB-CONDITIONAL · first introduced in Math 39
- I Determine whether events are independent or dependent.
  SK-PROB-INDEPENDENCE

M39-PRO-007

I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.

OpenStax: **Supplemental resources:** 11.7 Probability

SUPPORTING SKILLS

- I Determine whether events are overlapping or disjoint.
  SK-PROB-DISJOINT
- I Calculate the probability of a compound or event.
  SK-PROB-OR

M39-PRO-008

I can compare experimental and theoretical probability and explain the law of large numbers.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- D Calculate probability from observed relative frequency.
  SK-PROB-EMPIRICAL · first introduced in Math 39
- I Explain how experimental probability stabilizes with many trials.
  SK-PROB-LARGE-NUMBERS
- D Calculate theoretical probability for equally likely outcomes.
  SK-PROB-THEORETICAL · first introduced in Math 39

Expected Value

EXTENSION

OpenStax coverage: No mapped OpenStax section

M39-EV-001

I can calculate the expected value of a discrete probability situation.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I — Read expected-value notation and recall the meanings of outcome, probability, payoff, and long-run average.
  SK-PROB-EXPECTED-NOTATION
- I — Calculate the expected value of a discrete probability situation.
  SK-PROB-EXPECTED-VALUE

M39-EV-002

I can interpret expected value to evaluate a game, risk, or long-run decision.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I — Interpret expected value as a long-run average for decision-making.
  SK-PROB-EXPECTED-INTERPRET
- D — Calculate the expected value of a discrete probability situation.
  SK-PROB-EXPECTED-VALUE · first introduced in Math 39

Linearization of Nonlinear Data

EXTENSION

OpenStax coverage: No mapped OpenStax section

M39-POW-008

I can linearize power or exponential data and translate between the linearized and original models.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I Transform exponential data so that a linear model can be fit to its logarithms.
  SK-REG-LOG-LINEARIZE
- I Transform power data so that a linear model can be fit on logarithmic axes.
  SK-REG-LOGLOG-LINEARIZE
- I Translate a linearized regression equation back to its original nonlinear form.
  SK-REG-UNLINEARIZE